

One Rational Number, and a Stress-Test of the Inverted-Black-Hole Null

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Addendum to “The Inverted Black Hole: Why the de Sitter–MOND Acceleration Scale Is Uniquely Cosmic” (DOI 10.5281/zenodo.20947913) and “One Free Number” (DOI 10.5281/zenodo.20938891). All numbers are printed by committed scripts named per section, in the repository linked below.

Abstract

Three results, one of them a correction of presentation in the parent papers.

First, the coefficient reduces to a single rational. The parent note writes $a_0 = cH_\Lambda/Z$ with $Z = \sqrt{(32\pi/3)}$, factored as “the Einstein coupling 8π times the Friedmann 3, with a single free outside factor of 2.” That factorisation is now exact and complete: writing 32π as $8\pi/\kappa^2$ and substituting $\Lambda = 8\pi G\rho_\Lambda/c^2$, **every π , the 32 and the 3 all cancel**, and

$$a_0 = \kappa \cdot c \cdot \sqrt{(G \rho_\Lambda)}, \text{ exactly, with } \kappa = 1/2.$$

So “ $32\pi/3$ ” is not a compound of three geometric factors — it is an artifact of routing one statement through Λ and Einstein’s 8π . The framework’s entire dimensionless content is the single rational $\kappa = 1/2$, and the honest form of the open problem is “why $\kappa = 1/2$ ”, not “why $32\pi/3$ ”. This strengthens the *One Free Number* thesis: the number is not merely single, it is rational, and the transcendental $\sqrt{\pi}$ that the parent papers carry belongs to **general relativity**, not to this framework.

Second, the inverted-black-hole null survives a deliberate attack, for a stronger reason than the parent note gave. That note argues the universal crossover $r_{\text{cross}} = \sqrt{Z} \cdot r_s = 2.406 r_s$ self-cancels into exact GR via (a) mass-independence and (b) the free-fall/Hartle–Hawking theorem. Argument (b) protects only *geodesic* observers, which is a real gap: a **static** observer at radius r has proper acceleration $a = (GM/r^2)/\sqrt{(1 - r_s/r)}$, which is non-zero and diverges at the horizon. We evaluate the framework’s own kernel for exactly that observer at exactly r_{cross} . The fractional deviation from Newtonian inertia is

hole	$a_{\text{proper at } r_{\text{cross}}} \text{ (m s}^{-2}\text{)}$	$v - 1 \text{ (cosmic } a_0\text{)}$
stellar, $10 M_\odot$	3.438e+11	1.36e-22
Sgr A*	7.995e+05	5.85e-17
M87*	5.289e+02	8.85e-14

against a pre-declared interest threshold of $1e-6$. The worst case is **10^7 times below** what ngEHT or LISA could notice. A tidal channel fares worse still: every tidal scale at r_{cross} sits 10–21 orders *above* a_0 . So the null is confirmed on the non-geodesic branch too, and r_{cross} remains a prediction of **exactly-GR** shadows, ISCO frequencies, ringdown spectra and inspiral waveforms.

Third, and stated against interest, the parent note carries a presentational risk that a referee will find. It explicitly rejects the objection “ $g \gg a_0$ near a horizon, so the effect vanishes,” on the correct ground that the formal dual $a_0_{\text{BH}} = c^4/4GMZ$ scales up identically, keeping a/a_0_{BH} of order unity down to the horizon — indeed we find $v - 1 = 0.3283$ at r_{cross} , **identical for all three holes**, a clean exhibition of the mass-independence. But the framework’s *operative* scale is the **cosmic** a_0 — its own §1.2 states that inertia responds to the bath of the inverted cosmic horizon — and at that scale the effect *does* vanish, by ~ 22 orders. **Both statements are true of different scales, and they differ by 3.7×10^{12} at fixed radius.** The rejection is sound as written but reads as a claim of a real strong-field signal. One sentence fixes it, and this addendum supplies it.

1. The reduction: $a_0 = \kappa c \sqrt{(G\rho_\Lambda)}$

reviews/mi_kappa_spectral_reduction_2026.py.

Write the normalisation with κ explicit, $a_0 = c^2 \sqrt{(\kappa^2 \Lambda / 8\pi)}$, so that $\kappa = 1/2$ reproduces the published $c^2 \sqrt{(\Lambda / 32\pi)}$. Substituting the definition of the dark-energy density, $\Lambda = 8\pi G\rho_\Lambda / c^2$:

$$a_0 = c^2 \sqrt{(\kappa^2 \cdot 8\pi G\rho_\Lambda / (8\pi c^2))} = \kappa c \sqrt{(G\rho_\Lambda)}$$

verified symbolically. With $\kappa = 1/2$ this is $a_0 = (c/2) \sqrt{(G\rho_\Lambda)}$. Two consequences.

(i) The open problem is one rational. $Z(\kappa) = \sqrt{(8\pi/3)}/\kappa$, so $32\pi/3 = (8\pi/3)/\kappa^2$. “Derive Z ” and “derive κ ” are the same problem, and the second is the honest statement. The 8π is Einstein’s and the 3 is Friedmann’s; neither is the framework’s.

(ii) The transcendence obstruction belongs to GR, not here. The standing obstruction — “ Z carries a transcendental $\sqrt{\pi}$ while flavour data is algebraic” — is correct about Z , but Z ’s $\sqrt{\pi}$ comes from Einstein’s 8π . **The framework’s own number is rational.** This matters for how the obstruction is cited: it constrains what can be built from a GR-derived vocabulary, not what this framework asserts.

2. Stress-testing the null

real_research/reviews/mi_bh_cancellation_stress_2026.py (all checks pass, exit 0).

Four attacks, each with thresholds fixed before evaluation ($1e-6$ = of interest to ngEHT/LISA; $1e-3$ = already excluded by existing strong-field data).

A1, the static observer — the one that mattered. The free-fall theorem covers geodesic observers only. A static observer is non-geodesic and its proper acceleration diverges at the horizon, so this is a genuine gap in argument (b). Evaluating the framework’s own $v(y) = \sqrt{(1+1/y)}$ at $y = a_{\text{proper}}/a_0$ gives the table above: worst case $8.85e-14$, i.e. 1.1×10^7 below the interest threshold. **The gap is real and the conclusion survives it.**

A2, which scale is operative. At fixed radius the two readings differ by 3.7×10^{12} . This is not a matter of convention: the framework’s physical claim fixes the cosmic scale, and the dual is a formal exercise. §3 below states the consequence.

A3, the tidal channel. The kernel argument takes $\square_u u$, the second derivative along the worldline, not $|a|$. Being maximally generous (body size $L = r$) gives $a_{\text{tid}} \sim c^2 r_s / r^2$, the same order as g itself: 5.6×10^{21} , 1.3×10^{16} and 8.6×10^{12} times a_0 for the three holes. The tidal channel is *more* deeply switched off, not less.

A4, where the physics actually lives. Solving $a_{\text{proper}}(r) = a_0$ returns the a_0 shell, $r = \sqrt{(GM/a_0)}$: 0.122 pc for a stellar hole, 80 pc for Sgr A*, 3.1 kpc for M87* — that is **1.3×10^{11} , 1.9×10^8 and 5.0×10^6 Schwarzschild radii**. The framework meets a black hole parsecs to kiloparsecs outside it. $r_{\text{cross}} = 2.406 r_s$ is where the formal dual would switch on, never where the framework's physics does. (Both footings carried; alt-footing values in the script.)

3. The clarification the parent note needs

The parent note's rejection of " $g \gg a_0$ so the effect vanishes" is **correct about the dual and misleading about the framework**. Recommended single sentence:

The naïve objection fails for the formal dual a_0_{BH} , whose ratio a/a_0_{BH} stays $O(1)$ to the horizon; but the framework's operative scale is the cosmic a_0 , at which the deviation at r_{cross} is $\leq 8.85 \times 10^{-14}$ and the effect does vanish — so the strong-field prediction is a null either way, and the duality is structural rather than observational.

This subtracts a possible reading of a positive signal that was never claimed.

4. Two kept numbers from the de Sitter side

real_research/reviews/mi_bh_unravel_desitter_2026.py (22/22 pass).

- $T_{\text{dS}} / T_{\text{U}}(a_0) = Z$, exactly, canonical footing — the coefficient restated as a pure temperature ratio.
- **The a_0 shell of the maximal (Nariai) de Sitter black hole coincides with the de Sitter horizon:** $\sqrt{(Z/3\sqrt{3})} = 1.0555$ canonical, 0.9616 alt — within 6% of unity from pure numbers, no fit. Reported **with its ~2%-by-chance prior**: a structural coincidence, *not* a derivation of Z . Making it more would require deriving Z from the Schwarzschild–de Sitter double-root constant $3\sqrt{3}$ — the κ -forcing route, closed 2026-06-17 by ghost-freedom, unitarity and holography.

Also unchanged and not a framework result: every black hole de Sitter permits does evaporate, because the equilibrium mass $1.299 \times M_{\text{Nariai}}$ is forbidden, so $T_+ > T_c$ across 40 decades of mass. That is Gibbons & Hawking (1977) in plain GR + Λ . Modified inertia contributes **nothing**: at a stellar horizon $y = g/a_0 \sim 10^{23}$ and $v - 1 = 3.1 \times 10^{-23}$. **Black holes are the most purely general-relativistic objects this framework knows** — which is precisely why it survives the solar system.

5. What is not claimed

No new observable. No derivation of Z , κ , the response sign, or a_0 's value — all remain **postulated**. The strong-field axis stays a null whose discriminating power is against *metric-shifting* completions such as MOG, and not against this framework or against AeST, whose published black holes are stealth (Skordis & Złośnik, arXiv:2412.15395). Credit as in the parent papers: the interpolation

$v(y) = \sqrt{(1+1/y)}$ and the identity $g_{\text{obs}}^2 - g_{\text{bar}}^2 = a_0 g_{\text{bar}}$ are Milgrom (1999, *Phys. Lett. A* 253, 273) Eqs. (8)–(9); the a_0 – Λ tie is Milgrom’s and Smolin’s with a 2π coefficient, and **no priority is claimed for it**. What is ours is the coefficient cH_Λ/Z , the modified-inertia completion, and the results above.

References

Gibbons, G. W. & Hawking, S. W. 1977, *Phys. Rev. D* **15**, 2738. Nariai, H. 1950; Bousso, R. & Hawking, S. W. 1996 — the Nariai bound and near-extremal instability. Milgrom, M. 1999, *Phys. Lett. A* **253**, 273. Skordis, C. & Złośnik, T. 2024, arXiv:2412.15395 — stealth AeST black holes. Zimmerman, C. P. 2026, DOI 10.5281/zenodo.20947913 (parent), 10.5281/zenodo.20938891 (One Free Number).